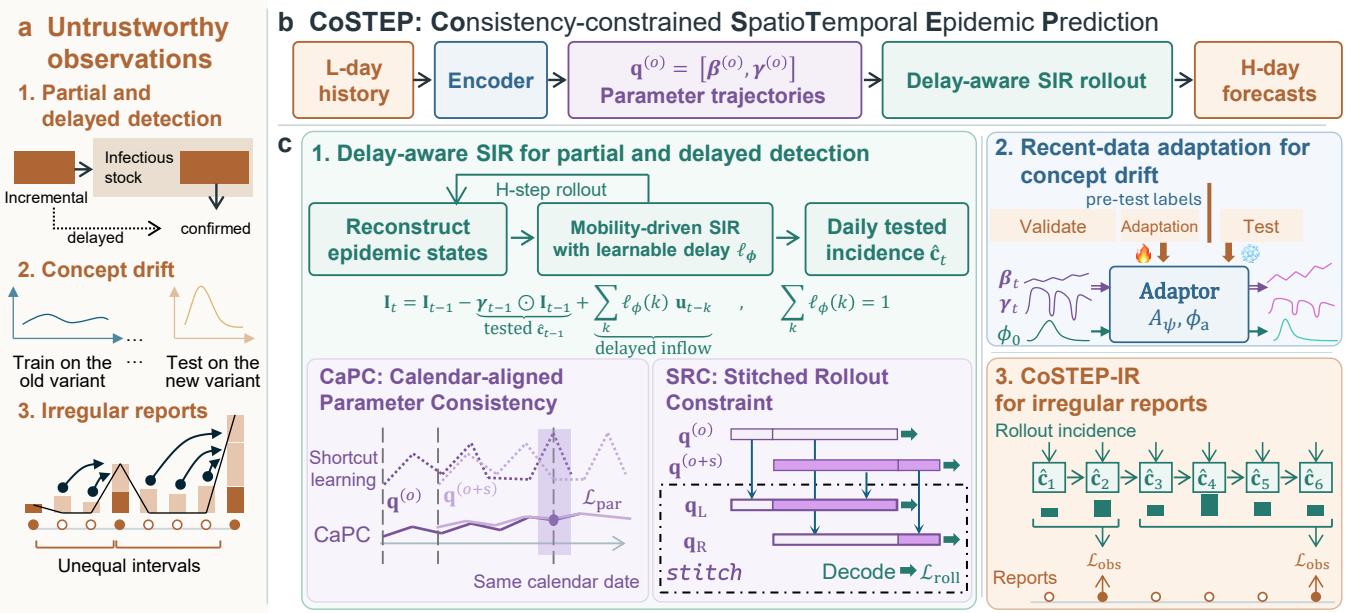

a Untrustworthy observations
1. Partial and delayed detection
Infectious stock
Incremental
delayed
confirmed
2. Concept drift
Train on the old variant
Test on the new variant
3. Irregular reports
Unequal intervals
b CoSTEP: Consistency-constrained SpatioTemporal Epidemic Prediction
L-day history
Encoder
$\mathbf{q}^{(o)} = [\boldsymbol{\beta}^{(o)}, \boldsymbol{\gamma}^{(o)}]$
Parameter trajectories
Delay-aware SIR rollout
H-day forecasts
c
1. Delay-aware SIR for partial and delayed detection
H-step rollout
Reconstruct epidemic states
Mobility-driven SIR with learnable delay ℓ_ϕ
Daily tested incidence $\hat{\mathbf{c}}_t$
$\mathbf{I}_t = \mathbf{I}_{t-1} - \underbrace{\boldsymbol{\gamma}_{t-1} \odot \mathbf{I}_{t-1}}_{\text{tested } \hat{\mathbf{c}}_{t-1}} + \underbrace{\sum_k \ell_\phi(k)\, \mathbf{u}_{t-k}}_{\text{delayed inflow}}$, $\sum_k \ell_\phi(k) = 1$
CaPC: Calendar-aligned Parameter Consistency
Shortcut learning
CaPC
$\mathbf{q}^{(o)}$
$\mathbf{q}^{(o+s)}$
$\mathcal{L}_{\text{par}}$
Same calendar date
SRC: Stitched Rollout Constraint
$\mathbf{q}^{(o)}$
$\mathbf{q}^{(o+s)}$
$\mathbf{q}_L$
$\mathbf{q}_R$
stitch
Decode
$\mathcal{L}_{\text{roll}}$
2. Recent-data adaptation for concept drift
pre-test labels
Validate
Adaptation
Test
$\boldsymbol{\beta}_t$
$\boldsymbol{\gamma}_t$
$\boldsymbol{\phi}_0$
Adaptor
A_ψ, ϕ_a
3. CoSTEP-IR for irregular reports
Rollout incidence
$\hat{\mathbf{c}}_1$ $\hat{\mathbf{c}}_2$ $\hat{\mathbf{c}}_3$ $\hat{\mathbf{c}}_4$ $\hat{\mathbf{c}}_5$ $\hat{\mathbf{c}}_6$
$\mathcal{L}_{\text{obs}}$
$\mathcal{L}_{\text{obs}}$
Reports